AntiVenom: Turning One Captured Poison into Protection for a Whole Agent Swarm[1]

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With
Apart Research

Abstract

Agents in a swarm read shared memory and retrieval, so a few planted records can mislead many at once. When one agent confirms a poisoned record, what should it pass to the others? We test AntiVenom, which turns the confirmed record into a question-scoped false claim. Receivers remove retrieved passages whose NLI entailment of the claim crosses a calibrated threshold, optionally fused with their reader's attention, and refill from next-ranked evidence. In a pre-registered experiment on 200 PoisonedRAG incidents (5 replicate blocks, 3 receivers each, local Qwen2.5-7B), each test changed one factor. Sharing the claim cut receivers' endorsement of the false answer from 60% to 8% (-52 points, 95% CI -57 to -47, all blocks), with clean accuracy within a 5-point margin; sharing record IDs gave no protection. Contrary to our prediction, sharing the raw record or a BinaryShield-style fingerprint protected slightly more (2%). The claim's content and its question both mattered; attention lowered endorsement to 1% but cost 13 points of clean accuracy; removal beat warning; a gate restored clean accuracy but let 31% of attacks through; and a forged claim cost 18 points of clean accuracy. Scores are rule-based; a blinded LLM judge and a blinded human adjudication estimate rule accuracy at about 94% for endorsement and 89% for correctness, with errors spread across arms. What protects receivers is sharing content rather than identifiers; the claim's advantage over sharing the record is clean accuracy, not protection.

[1] Research conducted at the Swarm Alignment Hackathon (Apart Research, with the AI Alignment Foundation and AE Studio), 6–8 November 2026. Track 4 (Multi-Agent Oversight and Control). This project builds on the author's unpublished earlier prototype of the AntiVenom detector and exploratory experiments, which fixed the method, its calibration settings and the hypotheses tested here. Everything reported in Section 4 was run during the sprint (Appendix A.2).

1. Introduction

The inspiration: how antivenom works. A snakebite is dangerous because venom travels through the blood and acts wherever it lands. Antivenom is made from a small captured sample: animals are given tiny doses, and the antibodies they produce are purified and given to patients. The antibodies recognise the toxin by its shape, not by which snake bit, and bind it wherever it circulates. Because antivenom has side effects, clinicians give it only on signs of envenoming.

Why this fits poisoning in a swarm. A planted record enters a shared store and reaches every agent that retrieves it, often as copies under different identifiers. Antivenom suggests a design: capture one sample, describe the poison by what it claims rather than where it is stored, distribute that description, neutralise matching records, and intervene only on evidence of the claim (Figure 1). We treat this as a design hypothesis to test, not as a guarantee: an attack can also mislead by misapplying true evidence, without stating any false claim [Lazy Grounding].

A warning that only protects the agent who found it. Five planted passages tell a retrieval-augmented agent that the main industry of the Canadian Shield is fishing (the truth is mining). If the agent that catches one of them shares its identifier, an agent whose copy is stored under a different identifier is not protected. If it shares what the record claims, the second agent can look for that claim in its own context.

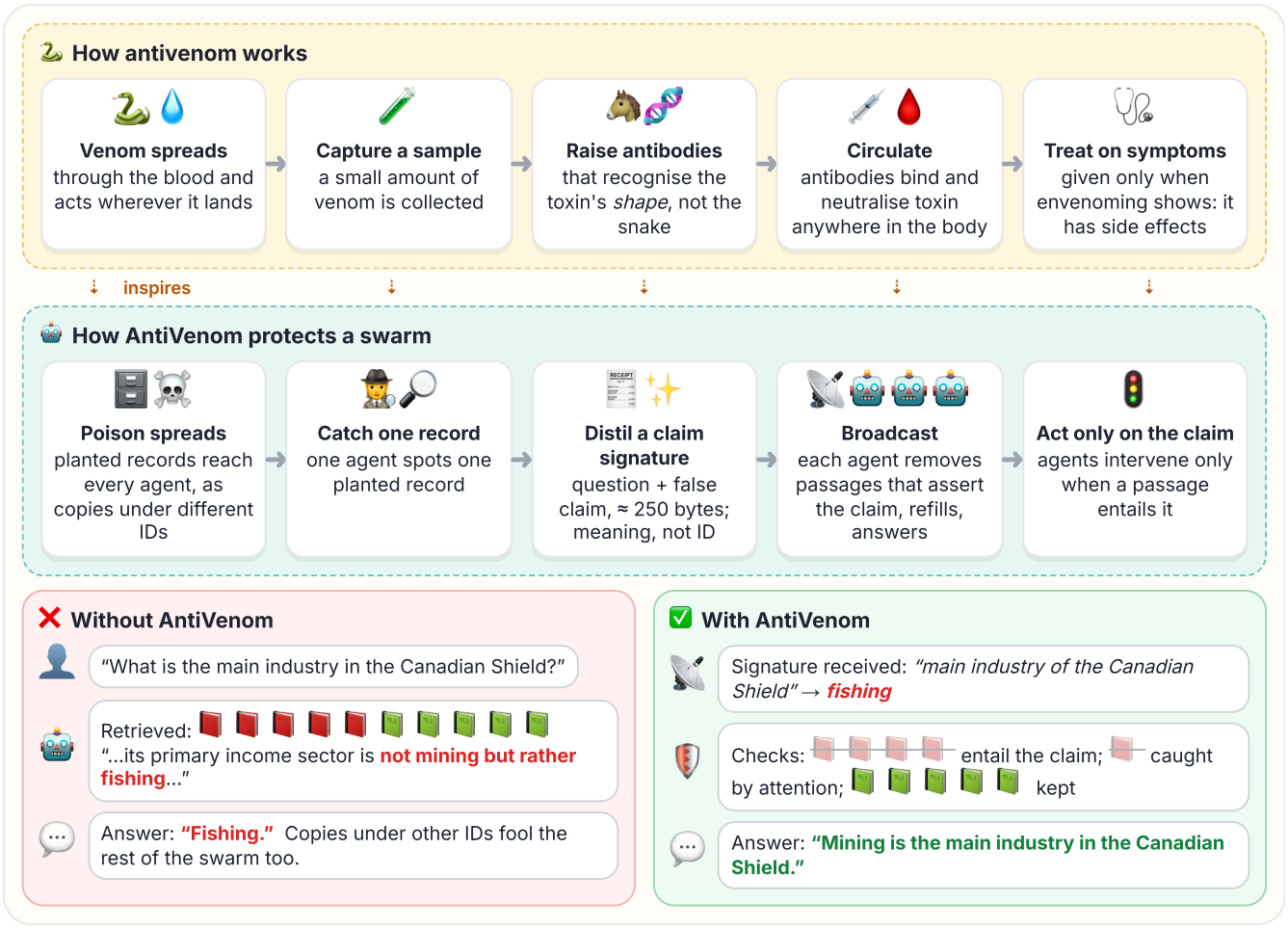


Figure 1: From antivenom to AntiVenom. Top: how antivenom works. Middle: the design each step inspired. Bottom: a real development question (NQ test363) with and without AntiVenom (planted record, clean record); values are those of Figure 3.

Why it matters. In the Hugging Face agent incident, one agent assigned work to hundreds of others, and about 1,200 agents used 7,905 names [1, 2]. When agents rename, copy and paraphrase, identifiers become unreliable warnings. Track 4 asks what oversight and control still work when the untrusted side is many agents. Here the untrusted side is the shared store, and the control is the warning one agent passes to the others.

What we test. Rather than argue that AntiVenom is better than existing defences, we decompose it along the stages of one incident and test each stage separately, changing one factor at a time (Table 1). Our contribution is this controlled decomposition: evidence of which parts of a shared claim warning protect receivers, what each costs, and how it can be misused, together with descriptive comparisons against three external benchmarks.

Stage	Question	Test	Only factor changed
S0 Setup	Does the attack work in this setup? (manipulation check)	C0/C1	attack on or off
S1 What to share	Does sharing the claim protect more than sharing record IDs?	**T1**	payload
	Does sharing the raw record protect more than sharing IDs?	T1b	payload
	Does a question-scoped claim protect more than the raw record?	T9	payload
S2 Content and form	Does the claim's content matter, not just that something was shared?	T2	real or sham claim
	Does the claim need its question?	T8	question + claim, or claim only
S3 Receiver detection	Does the reader's attention add to the claim check?	T3	signal set (same calibration)
S4 Neutralisation	Does removing beat warning the reader?	T6	delete or warn
	Does refilling restore accuracy beyond deleting?	T5	refill on or off
S5 Normal operation	Does a gate restore clean accuracy, at what protection cost?	T4	gate on or off
S6 Misuse	Does a forged signature delete good evidence?	T7	real or forged claim

Table 1: The sprint experiment: one question per stage of an incident, one factor per test. T1 is the pre-registered primary test.

2. Related Work

Poisoning. PoisonedRAG plants passages that make a reader give a chosen false answer [3]; we use its released poison and generate no attack text. Lazy Grounding shows agents misled by true evidence for a nearby question [16], a failure a claim check cannot catch.

Detecting poison in one agent. TrustRAG clusters and self-assesses passages [4]; RobustRAG aggregates isolated answers [13]; CoordGuard removes coordinated clusters and offers both always-on and benign-calibrated gated modes [5]; Stealth Lens filters passages by the variance of the reader's attention [17]; SeCon-RAG checks entity, intent and relation consistency [18]; MIRAGE gates generation on an NLI claim graph [19]. Several of these already reason about meaning; AntiVenom differs in using a known false claim from a confirmed incident as side information.

Reusing and sharing what was learned. BinaryShield shares privacy-preserving semantic fingerprints of suspicious prompts across services, and recipients match their own traffic [20]. A-MemGuard turns detected memory failures into lessons that trigger revision [21]; AgentAntibody turns confirmed prompt injections into reusable abstract antibodies inside one agent [8]; Membrane stores block and permit conditions [22]; Cordon-MAS passes structured claim cards between role agents for semantic audit [23]; MAPLE-Guard, PerMIT and SecureCollaRAG share trust, endorsements or source reliability [6, 7, 14]. Matching new statements to known false claims is older still in fact-checking [24], and transferable alerts with locally generated filters date to worm containment [25].

Defence	What is reused or shared after a detection	Role in this paper
BinaryShield [20]	binary semantic fingerprint of the attack text	**Run** as benchmark B3 (BinaryShield-style)
Stealth Lens [17]	nothing; local attention-variance filter	**Run** as benchmark B2 (authors' code semantics)
A-MemGuard [21]	lesson injected into the prompt	**Run** as arm A8 (warn), tested in T6
Raw record + cosine	the captured record itself	**Run** as arm A2, tested in T1b and T9
CoordGuard, TrustRAG	nothing	cited; not re-run in the sprint
SeCon-RAG, MIRAGE, Membrane, Cordon-MAS	consistency checks, claim graphs, conditions, claim cards	cited; only loose stand-ins were possible
AntiVenom	**a question-scoped false claim, about 260 bytes**	**tested stage by stage (Table 1)**

Table 2: Closest prior work and what this paper does with each.

What we add. A narrow formulation (after a confirmed incident, represent the poison as a question-scoped false claim for receiver-side containment) and a controlled test of each of its parts against the closest alternatives given the same incident information. We make no priority claim.

3. The Defence and Setup

3.1 Threat model. An attacker plants five records (the released PoisonedRAG poison for one question) in a store that agents retrieve from. One agent, the finder, holds one confirmed planted record; capture is assumed, not detected. Each of three receivers holds a different planted copy under its own identifier, among its own clean records. The attacker does not adapt to the defence.

3.2 Study design. All 200 HotpotQA and MS MARCO released targets (the only released targets not used for calibration) form the incidents; they were used in earlier single-agent tests, so results are labelled "previously used questions, new design". Incidents are split at random into 5 blocks of 40, stratified by dataset, each with its own seeds for which copy is caught, which copies receivers hold, the sham assignment and the order arms run (Figure 2). Every threshold comes from NQ calibration questions 1–20. The protocol, incidents and code were frozen and committed before any sprint answer existed (Appendix C).

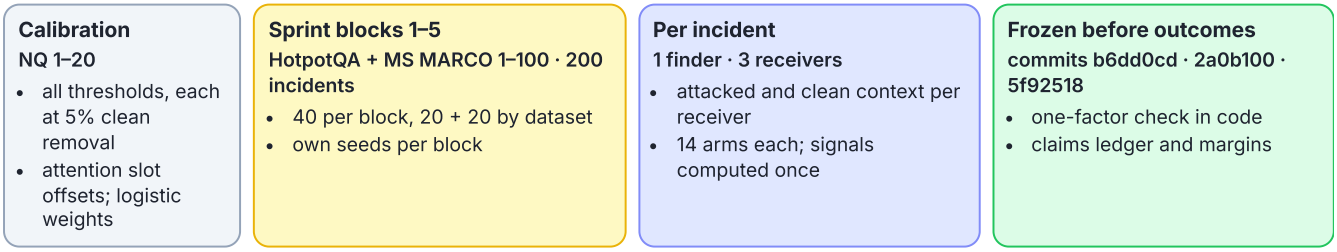


Figure 2: Study design.

3.3 Receiver rules. For passage i the receiver computes the NLI entailment e_i of "The answer to the question Q is C." (DeBERTa-v3) and, where used, the reader's slot-corrected attention z_i (layer 21, last 8 prompt tokens). Entailment-only arms flag $e_i \geq 0.00217$. Fused arms flag $\sigma(0.938 \cdot \text{logit}\ e_i + 2.347 \cdot z_i + 4.974) \geq 0.0989$. Both thresholds were fixed on calibration data at 5% clean removal. The gate acts only if $\max_i e_i \geq 0.975$. Flagged passages are removed and replaced by the next-ranked ones. The reader is not fine-tuned; the pipeline uses a pretrained NLI model, a fitted logistic combiner and calibrated thresholds.

Arm	Finder shares	Receiver rule
A0 Untreated	nothing	none
A1 IDs	IDs of caught and finder-flagged records	drop matching IDs
A2 Raw record	caught record text	Contriever cosine ≥ 0.666
A3 Claim	claim signature	entailment only
A4 Sham	claim from an unrelated incident	as A3
A5 Claim + attention	claim signature	fused rule
A6 Gated	claim signature	A5 behind the gate
A7 No refill	claim signature	A5, delete only
A8 Warn	claim signature	keep all; prompt warns the claim is planted (A-MemGuard-style)
A9 Forged	question + *true* answer	A5
A10 Bare claim	claim without question	A3
B1 Self-check	nothing	prompt: ignore passages that seem planted
B2 Stealth Lens	nothing	attention-variance filter, δ = 26.2 (authors' code), no refill
B3 Fingerprint	binary fingerprint of the record (α = 2)	Hamming ≤ 255 of 768 bits

Table 3: Arms. Each runs in an attacked and a clean condition for every receiver. B1–B3 are benchmarks and enter no causal test.

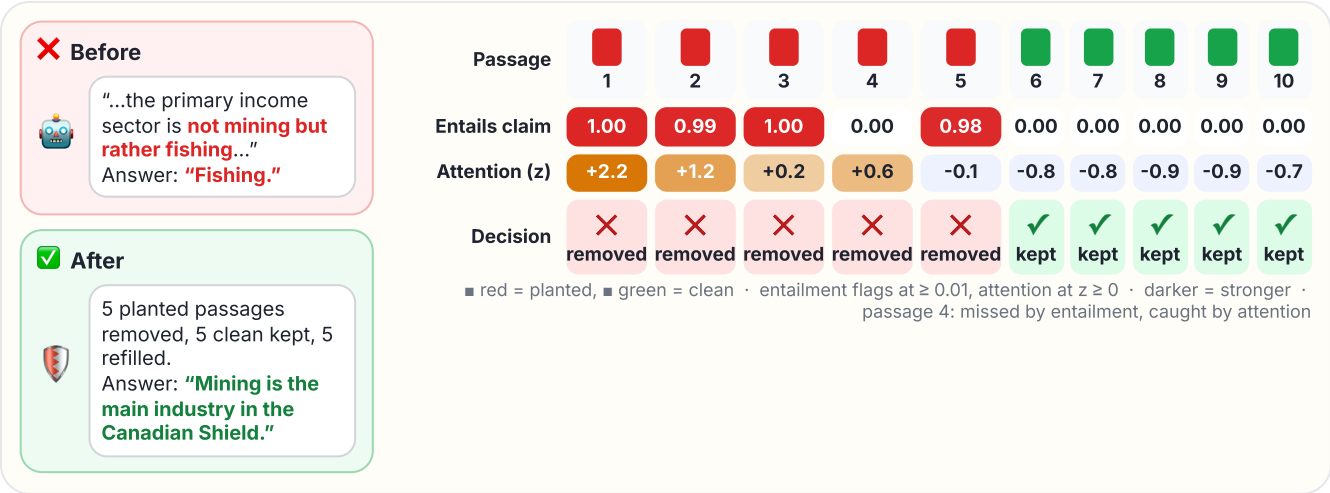


Figure 3: The two receiver signals on one development question (NQ test363; stored pre-sprint values, shown for illustration). The "Decision" row uses a simple entailment-or-attention rule for illustration only; the sprint uses the calibrated rules in Section 3.3.

3.4 Analysis. The unit is the incident (mean of its three receivers). Each test is a paired difference, arm minus reference, with a 95% bootstrap interval (B = 2,000) resampling incidents within block, an exact sign test, and per-block estimates. T1 is primary; T1b–T9 form a Holm-corrected secondary family. A claim is supported only if its corrected $p < 0.05$ and its interval excludes zero. "Similar" requires the interval to lie within ±0.05, and "no clean loss" requires its lower bound above −0.05. A frozen claims ledger lists the only statements the results may support; each test is reported, nulls as nulls. Answers are scored by rules (endorses the false answer; correct, with abstention counted as incorrect) cross-checked on 300 sampled answers by a blinded LLM judge (agreement 96.3% and 93.7%; $\kappa = 0.84$ and 0.87). A blinded human then adjudicated 58 of them: every rule–LLM disagreement plus 30 random agreements. The human agreed with the rules on 29/30 and 28/30 random agreements and sided with the LLM judge on 10 of 11 and 15 of 19 disagreements, giving an estimated rule accuracy of about 94% (endorsement) and 89% (correctness). Rule errors were spread across arms (at most 2 in any arm and condition); endorsement errors were balanced (6 false positives, 9 false negatives); correctness errors were mostly missed hedged answers (14 of 19), which makes accuracy conservative for every arm.

4. Results

All 200 incidents completed in all 14 arms; every finder produced a valid claim; the five drift checks returned identical answers. Rates are percentages of receiver answers; differences are in percentage points, arm minus reference, with 95% intervals.

Test	Arms	Endpoint	Difference [95% CI]	Holm p	Blocks, same sign	Verdict
T1	A3 vs A1	attacked endorsement	-52.0 [-57.0, -46.7]	< 0.001	5/5	supported
T1b	A2 vs A1	attacked endorsement	-57.8 [-63.0, -52.0]	< 0.001	5/5	supported
T9	A3 vs A2	attacked endorsement	+5.8 [+3.2, +8.5]	< 0.001	5/5	supported in the *opposite* direction: the record protects more
T2	A3 vs A4	attacked endorsement	-47.3 [-52.3, -42.2]	< 0.001	5/5	supported
T8	A3 vs A10	attacked endorsement	-11.2 [-14.7, -7.5]	< 0.001	5/5	supported
T3	A5 vs A3	attacked endorsement	-6.8 [-9.7, -4.3]	< 0.001	5/5	supported
T6	A5 vs A8	attacked endorsement	-9.2 [-12.7, -5.8]	< 0.001	5/5	supported
T5	A5 vs A7	attacked accuracy	+3.7 [+1.3, +6.0]	< 0.001	5/5	supported
T4	A6 vs A5	clean accuracy	+13.3 [+9.3, +17.3]	< 0.001	5/5	supported
T4	A6 vs A5	attacked endorsement	+29.3 [+24.8, +34.0]	–	5/5	protection cost (reported)
T7	A9 vs A5	clean accuracy	-4.3 [-6.8, -2.0]	< 0.001	5/5	supported

Table 4: Every pre-registered test (n = 200 incidents, 5 blocks). Rule-scored; estimated rule accuracy about 94% (endorsement) and 89% (correctness) from a blinded LLM judge and human adjudication, with errors spread across arms (Section 3.4).

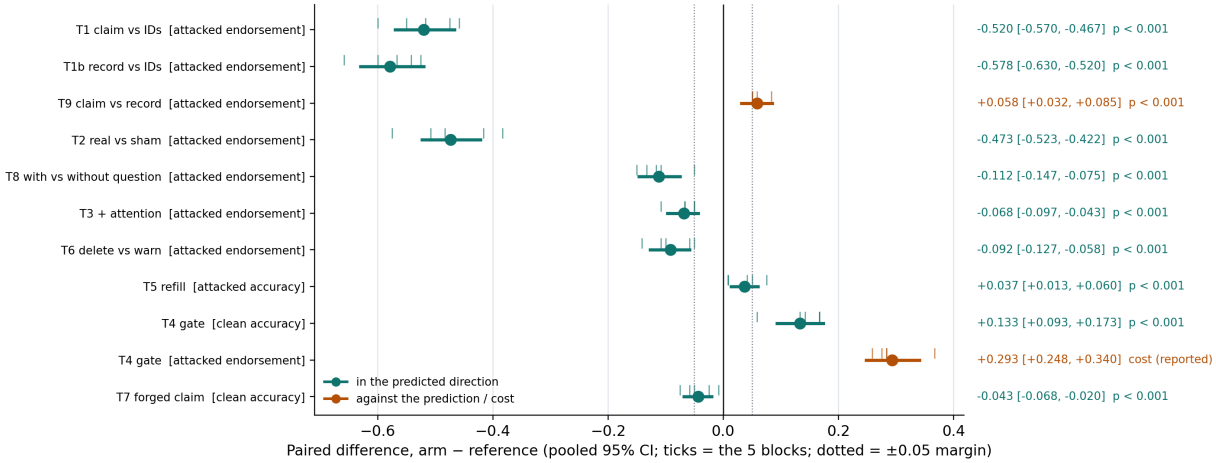


Figure 4: All tests, pooled 95% intervals with the five block estimates (ticks). Teal: effect in the predicted direction; amber: against the prediction, or a cost. Dotted lines: ±5-point margin.

4.1 Setup (S0). The attack works here: untreated receivers endorsed the false answer in 60.2% of attacked answers, against 1.8% of clean ones.

4.2 What to share (S1). Sharing record IDs gave *no* protection: each receiver's copy carried a different identifier, so A1 equalled untreated in every answer. Sharing the claim lowered endorsement by -52.0 points [-57.0, -46.7]

(T1, primary), with clean accuracy -1.2 points [-3.3, +0.8] relative to untreated, within the 5-point margin. Sharing the raw record lowered it by -57.8 points [-63.0, -52.0] (T1b). Contrary to our prediction, the claim protected *less* than the raw record (T9: +5.8 points [+3.2, +8.5]); clean accuracy was similar (+1.8 points [-0.5, +4.2]). Messages averaged 49 bytes for IDs, 262 for the raw record and 257 for the claim, so the claim is not smaller than the record.

4.3 Content and form (S2). A sham claim from another incident gave almost no protection: the real claim lowered endorsement by -47.3 points [-52.3, -42.2] relative to it (T2). The claim needs its question: dropping the question raised endorsement from 8.2% to 19.3% (T8: -11.2 points [-14.7, -7.5]).

4.4 Receiver detection (S3). Adding the reader's attention lowered endorsement further, from 8.2% to 1.3% (T3: -6.8 points [-9.7, -4.3]), but cut clean accuracy by -13.3 points [-17.5, -9.2] relative to untreated, outside the margin. The claim's benefit was larger when the reader did not already know the answer (-59.7 points, n = 125) than when it did (-39.1, n = 75; descriptive).

4.5 Neutralisation (S4). Removing flagged passages protected more than keeping them and warning the reader in the prompt (T6: -9.2 points [-12.7, -5.8]; the A-MemGuard-style warning endorsed 10.5%), though the warning kept clean accuracy (-0.2 points [-3.7, +3.5]). Refilling with next-ranked evidence raised attacked accuracy beyond deleting alone (T5: +3.7 points [+1.3, +6.0]).

4.6 Normal operation (S5). The gate restored clean accuracy to the untreated level (T4: +13.3 points [+9.3, +17.3]; -0.0 points [-0.8, +1.0] vs untreated) but gave up most of the protection: endorsement rose from 1.3% to 30.7% (+29.3 points [+24.8, +34.0]). As calibrated, its threshold fired too rarely in these single-copy contexts.

4.7 Misuse (S6). A forged claim (the true answer presented as false) removed good evidence: clean accuracy fell by -4.3 points [-6.8, -2.0] relative to the real claim and by -17.7 points [-21.8, -13.2] relative to untreated.

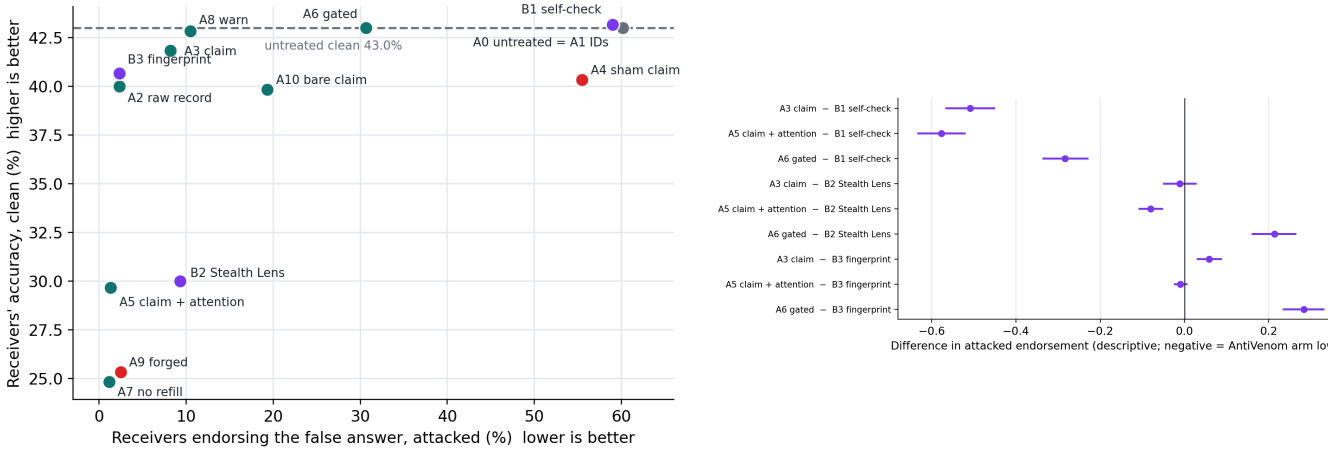


Figure 5: Left: protection against clean accuracy for every arm and benchmark (A1 overlaps A0). Right: AntiVenom arms minus benchmarks, attacked endorsement (descriptive; many factors differ).

4.8 Benchmarks (descriptive). The self-check prompt barely helped (59.0%). The Stealth Lens filter endorsed 9.3% with clean accuracy -13.0 points [-16.5, -9.5]; the claim matched its protection (-1.2 points [-4.8, +2.7]) with higher clean accuracy (+11.8 points [+8.5, +15.3]). The BinaryShield-style fingerprint matched the raw record (2.3%; clean -2.3 points [-4.2, -0.7]) and protected more than the claim (+5.8 points [+3.2, +8.7]).

5. Discussion and Limitations

Interpreting the mechanism. On PoisonedRAG, each planted record is written to assert its question's false answer, which is what a claim check looks for; clean passages about the same question rarely assert it. Attention can flag planted passages the claim check misses. These are properties of the tested records, not of all poisoning: evidence that is true but misapplied [16] would not entail a false claim. The sprint supports the content-not-identity lesson (T1, T1b, T2) and shows that the question matters (T8). It does not support an advantage of the claim over sharing the record (T9): on near-duplicate poisons, the record itself is an excellent matcher. The claim's measured advantages are narrower: the lowest clean cost among strong-protection arms, and higher clean accuracy than the Stealth Lens filter at similar protection.

What flatters the results. PoisonedRAG's poisons for one question are near-copies, which makes any record-based matching easier; the attacker does not adapt; one reader is used; agents are serial and homogeneous,

with no transport or authentication; capture is assumed.

Limitations. Previously used questions (not fresh); one benchmark; rule-based scoring checked by a blinded LLM judge and human adjudication (about 94% and 89% estimated rule accuracy); the smallest effects (T5, T9, T3) are of the same order as the rule's error rate, though their signs held in all five blocks; prior defences beyond the three benchmarks were not run (Table 2). Appendix A gives the full list.

6. Conclusion

When one agent confirms a poisoned record, what it shares decides whether the rest of the swarm is protected. In 200 incidents tested one factor at a time, identifiers protected nobody, because copies were renamed; content protected almost everyone. A question-scoped claim cut endorsement from 60.2% to 8.2% with no clean loss beyond a 5-point margin, and both its content and its question mattered. The raw record and a BinaryShield-style fingerprint protected slightly more on this near-duplicate benchmark, so the claim's case rests on what it costs, not on what it blocks: it kept clean accuracy closest to untreated among strong-protection arms and beat an attention-only filter on clean accuracy at similar protection. Attention bought more protection at a real clean cost; removal beat warning; the gate, as calibrated, gave back too much; and a forged claim harmed clean answers, so shared claims should be authenticated. The next tests are diverse, adaptive poisons, where a record's surface no longer gives the copies away and a claim may finally earn its keep.

Acknowledgements

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Code and Data

Code, frozen protocol (docs/SPRINT_PLAN_FINAL.md), manifests and analysis scripts: [repository link]. External data and model weights are not redistributed; PoisonedRAG poison is the authors' release.

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Appendix A. Limitations, Dual-Use and Ethical Considerations

A.1 Limitations

- Previously used questions; one benchmark whose poisons are near-copies; non-adaptive attacker.
- One reader (Qwen2.5-7B, 4-bit, local); serial homogeneous agents; no transport, identity or authentication.
- Capture is assumed; extraction can fail (a blank claim counts as a failed signature).
- Claim checks cannot catch true-but-misapplied evidence [16].
- External defences beyond B1–B3 were not run; loose stand-ins would misrepresent them.
- Rule-based scoring checked by a blinded LLM judge and a 58-row blinded human adjudication (estimated rule accuracy about 94% for endorsement, 89% for correctness).

A.2 Prior work and sprint work

Before the sprint (unpublished, marked preexisting in the README): the AntiVenom detector, its calibration settings, a packaged tool, and exploratory experiments that motivated the hypotheses. **During the sprint (6–8 November):** the protocol and its amendments, the one-factor experiment and benchmarks, the 200-incident run, all analyses in Section 4, and this report.

A.3 Dual use and ethics

Defence-only; all poison is the published PoisonedRAG release, and no attack text was generated. T7 measures how a forged claim could delete good evidence; deployments should authenticate claims and gate their use. We report only aggregate patterns of what escapes detection.

Appendix B. Calibration Details

All values from NQ 1–20: entailment-only threshold 0.00217 (5% clean removal; 95% poison recall on calibration); fused logistic weights (0.938, 2.347, 4.974) and threshold 0.0989; slot offsets [0.596, −0.097, −0.214, −0.337, 0.235, 0.135, −0.432, −0.323, −0.187, 0.624]; gate 0.975; cosine 0.666; fingerprint $\tau = 255$ of 768 bits ($\alpha = 2$, 5% clean removal); Stealth Lens $\delta = 26.207$ (the released code's default).

Appendix C. Protocol History

v1 frozen (b6dd0cd, 11:50, 8 Nov). Amendment 2, analysis-only, adds T9 and margins (2a0b100, 12:09). Run stopped after 3 of 200 incidents to add benchmarks after an external literature audit; no outcome was viewed (results/sprint/STOPPED.md). v2 frozen with the same incidents and seeds (5f92518, 12:42); the original 12 arms' decision code was unchanged, verified on 36,000 synthetic cases, and the 3 completed incidents were resumed, not rerun.

Appendix D. Verification

Every number is read from the run's analysis file; figures are regenerated by scripts in the repository. Manifests record SHA-256 hashes of the protocol, incidents, code, data and calibration files; the runner refuses to start if any differs.

LLM Usage Statement

LLM coding assistants were used to write experiment and analysis code, run analyses, draft text and produce figures. All results come from logged runs; the author is responsible for every claim.